

Question: Did Guyana’s oil discovery increase GDP growth relative to CARICOM?

Countries: Barbados, Jamaica, Trinidad & Tobago, Suriname, Belize, and Grenada.
Model:

$$GDP_growth = \beta_0 + \beta_1 did + \text{Year FE} + \text{Country FE} + \epsilon$$

Did = treatment x post

Country FE: controls for differences within countries such as population size, industrial base, political stability, geography, etc.
Removes baseline differences within countries such as Jamaica growing differently than Trinidad & Tobago.

Year FE: Controls for differences over time.

Data

Panel Data Source & Initial Observation Preview							
Status Message: Panel dataset created successfully!							
File Destination: C:/Users/Ravin/Downloads/Guyana_CARICOM_GDP_long.csv							
Index	Country Name	Country Code	Year	GDP_growth (%)	treatment	post	did
0	Guyana	GUY	2000	-1.363804	1	0	0
1	Barbados	BRB	2000	4.447189	0	0	0
2	Jamaica	JAM	2000	0.878704	0	0	0
3	Trinidad and To- bago	TTO	2000	6.901360	0	0	0
4	Suriname	SUR	2000	2.099999	0	0	0

2. OLS Model Performance & Goodness-of-Fit Summary

Model Parameters		Fit Statistics & Information Criteria	
Dep. Variable:	GDP_growth	R-squared:	0.533
Model:	OLS	Adj. R-squared:	0.433
Method:	Least Squares	F-statistic:	5.313
No. Observations:	182	Prob (F-statistic):	9.35 × 10 ⁻¹³
Df Residuals:	149	Log-Likelihood:	-575.14
Df Model:	32	AIC:	1216.
Covariance Type:	nonrobust	BIC:	1322.

3. OLS Estimation Results & Coefficients

Variable / Factor	Coefficient	Std. Error	t-statistic	P > t	[95% Conf. Interval]
Intercept	3.1112	2.648	1.175	0.242	[-2.122, 8.344]
Year Fixed Effects (Baseline Year: 2000)					
C(Year)[T.2001]	-2.4993	3.370	-0.742	0.459	[-9.158, 4.159]
C(Year)[T.2002]	-0.9431	3.370	-0.280	0.780	[-7.601, 5.715]

Variable / Factor	Coefficient	Std. Error	t-statistic	$P > t $	[95% Conf. Interval]
C(Year)[T.2003]	2.1411	3.370	0.635	0.526	[-4.517, 8.800]
C(Year)[T.2004]	-0.7443	3.370	-0.221	0.825	[-7.403, 5.914]
C(Year)[T.2005]	-0.1413	3.370	-0.042	0.967	[-6.800, 6.517]
C(Year)[T.2006]	0.5036	3.370	0.149	0.881	[-6.155, 7.162]
C(Year)[T.2007]	-0.0087	3.370	-0.003	0.998	[-6.667, 6.650]
C(Year)[T.2008]	-3.0943	3.370	-0.918	0.360	[-9.753, 3.564]
C(Year)[T.2009]	-6.3259	3.370	-1.877	0.062	[-12.984, 0.333]
C(Year)[T.2010]	-3.7276	3.370	-1.106	0.270	[-10.386, 2.931]
C(Year)[T.2011]	-2.4357	3.370	-0.723	0.471	[-9.094, 4.223]
C(Year)[T.2012]	-2.2996	3.370	-0.682	0.496	[-8.958, 4.359]
C(Year)[T.2013]	-1.8581	3.370	-0.551	0.582	[-8.516, 4.800]
C(Year)[T.2014]	-1.6782	3.370	-0.498	0.619	[-8.337, 4.980]
C(Year)[T.2015]	-6.4545	3.392	-1.903	0.059	[-13.156, 0.248]
C(Year)[T.2016]	-7.4495	3.392	-2.196	0.030	[-14.151, -0.748]
C(Year)[T.2017]	-6.4890	3.392	-1.913	0.058	[-13.191, 0.213]
C(Year)[T.2018]	-5.0732	3.392	-1.496	0.137	[-11.775, 1.629]
C(Year)[T.2019]	-5.3474	3.392	-1.577	0.117	[-12.049, 1.355]
C(Year)[T.2020]	-12.2397	3.392	-3.609	0.000	[-18.942, -5.538]
C(Year)[T.2021]	0.2738	3.392	0.081	0.936	[-6.428, 6.976]
C(Year)[T.2022]	7.7348	3.392	2.281	0.024	[1.033, 14.437]
C(Year)[T.2023]	-0.5912	3.392	-0.174	0.862	[-7.293, 6.111]
C(Year)[T.2024]	0.7689	3.392	0.227	0.821	[-5.933, 7.471]
C(Year)[T.2025]	-3.0174	3.392	-0.890	0.375	[-9.719, 3.685]
Country Fixed Effects (Baseline Country: Barbados)					
C(Country_Name)[T.Belize]	2.4934	1.748	1.426	0.156	[-0.962, 5.948]
C(Country_Name)[T.Grenada]	1.6540	1.748	0.946	0.346	[-1.801, 5.109]
C(Country_Name)[T.Guyana]	1.0074	2.089	0.482	0.630	[-3.121, 5.136]
C(Country_Name)[T.Jamaica]	0.2571	1.748	0.147	0.883	[-3.198, 3.712]
C(Country_Name)[T.Suriname]	1.2595	1.748	0.720	0.472	[-2.195, 4.714]
C(Country_Name)[T.Trinidad and Tobago]	1.6424	1.748	0.939	0.349	[-1.812, 5.097]
Treatment Estimator (Difference-in-Differences Interaction)					
did	21.3116	2.703	7.885	0.000	[15.971, 26.653]

4. Model Residual Diagnostics & Policy Takeaway

Diagnostic Test	Value	Diagnostic Test	Value
Omnibus Test:	71.818	Durbin-Watson Stat:	2.189
Prob(Omnibus):	0.000	Jarque-Bera (JB) Stat:	590.958
Skewness:	1.210	Prob(JB):	4.73×10^{-129}
Kurtosis:	11.489	Condition No.:	28.7

The treatment variable was created to identify which country received the shock

Treatment = 1 for Guyana

Treatment = 0 for all other CARICOM countries

This is needed so that the model can distinguish Guyana from the control Group

Post variable

Needed to confirm when the shock happened

Post = 1 for years $\geq$ 2015

Post = 0 for years $<$ 2015

This allows the model to know the oil discovery happened in 2015

Did = treatment x post

- Guyana (treatment =1)
- After 2015 (post =1)

Did = 1 for Guyana alone after the oil discovery

And 0 for everyone else

Statistical Significance of Oil: Estimated coefficient $\beta_{did} = 21.3116$ ($t = 7.885$, $p < 0.001$) making it statistically significant. Post-discovery, Guyana saw an average additional **21.31** percentage point increase in real GDP growth relative to CARICOM control countries

Fit: The model accounts for 53.3% of the variance ($R^2 = 0.533$, adjusted $R^2 = 0.433$), it has a highly significant overall F-statistic($P = 9.35 \times 10^{-13} < 0.05$).